

Tutorial Questions (Set-V)

- Q1 (i) Let $f: [a, b] \rightarrow \mathbb{R}$ be a function and $x \in [a, b]$. Show that f is differentiable at x with derivative $f'(x) = \ell$ if and only if there exists a function $u(t)$ such that $\lim_{t \rightarrow x} u(t) = 0 = u(x)$ and

$$f(t) = f(x) + (t - x)(\ell + u(t)), \quad t \in [a, b].$$

- (ii) Let $f: [-1, 1] \rightarrow \mathbb{R}$ be given by $f(t) = \begin{cases} t^2 & \text{if } t \in \mathbb{Q} \cap [-1, 1], \\ 0 & \text{otherwise.} \end{cases}$ Show that f is differentiable at 0. Determine $u(t)$ for f . Further, show that f is not continuous at any $x \neq 0$ in $[-1, 1]$.

- (iii) Let $g(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$ Show that g is a differentiable function on $\mathbb{R}$ but the derivative g' is not continuous at 0.

- Q2 Let $a, c \in \mathbb{R}$ with $c > 0$. Let $h: [-1, 1] \rightarrow \mathbb{R}$ be a function defined by

$$h(t) = \begin{cases} |t|^a \sin(|t|^{-c}) & \text{if } t \neq 0, \\ 0 & \text{if } t = 0. \end{cases}$$

Prove the following statements.

- (i) h is continuous $\Leftrightarrow a > 0$. Further, $h'(0)$ exists $\Leftrightarrow a > 1$.
 - (ii) h' is bounded $\Leftrightarrow a \geq 1 + c$. Also, h' is continuous $\Leftrightarrow a > 1 + c$.
 - (iii) $h''(0)$ exists $\Leftrightarrow a > 2 + c$. Also, h'' is bounded $\Leftrightarrow a \geq 2 + 2c$.
 - (iv) h'' is continuous $\Leftrightarrow a > 2 + 2c$.
- Q3 (i) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $|f(x) - f(y)| \leq (x - y)^2, \forall x, y \in \mathbb{R}$. Show that f is differentiable on $\mathbb{R}$ with $f'(x) = 0 \forall x \in \mathbb{R}$. Use Lagrange's Mean-value theorem (MVT) to deduce that f is constant.
- (ii) Let $c_1, c_2, \dots, c_n$ be real numbers and $\sum_{k=0}^n \frac{c_k}{k+1} = 0$. Show that the polynomial $\sum_{k=0}^n c_k x^k$ has a root in $(0, 1)$. (Hint : Use MVT)
- (iii) Suppose $f: [-1, 1] \rightarrow \mathbb{R}$ is a 3-times differentiable function such that $f(-1) = 0, f(0) = 0, f(1) = 1$ and $f'(0) = 0$. Show that $\exists x \in (-1, 1)$ such that $f^{(3)}(x) \geq 3$. Note that for $f(t) = \frac{1}{2}(t^3 + t^2)$, we have $f^{(3)}(0) = 3$.
- (iv) Give an example of a real function which is n -times continuously differentiable but not $(n + 1)$ -times differentiable for $1 \leq n \leq 4$.

- Q4 (i) Use L'Hospital's rule to determine $\lim_{x \rightarrow 0+} \frac{\sin x}{\sqrt{x}}$, $\lim_{y \rightarrow \infty} \frac{y}{e^y}$ and $\lim_{x \rightarrow 0+} \frac{e^{-\frac{1}{x}}}{x}$.
- (ii) Let $f, g: [a, b] \rightarrow \mathbb{R}$ be continuous functions with $f(a) = g(a) = 0$ and $g(t) \neq 0 \forall t \in (a, b)$. Suppose f and g are differentiable at a and $g'(a) \neq 0$. Show that if $\lim_{t \rightarrow a+} \frac{f(t)}{g(t)}$ exists, then $\lim_{t \rightarrow a+} \frac{f(t)}{g(t)} = \frac{f'(a)}{g'(a)}$.
- (iii) Let $P(y)$ be a real-polynomial in y of degree $n \geq 1$. By repeated use of L'Hospital's rule, show that $\lim_{y \rightarrow \infty} \frac{P(y)}{e^y} = 0$. Hence, $\lim_{x \rightarrow 0+} P\left(\frac{1}{x}\right) e^{-\frac{1}{x}} = 0$.
- (iv) Suppose f is a real-differentiable function on an open interval $I = (a, b)$ and $x \in I$ such that $f''(x)$ exists. Show that

$$\lim_{h \rightarrow 0} \frac{f(x+h) + f(x-h) - 2f(x)}{h^2} = f''(x).$$

Show by an example that limit may exists even if $f''(x)$ does not.

- Q5 A function $f: (a, b) \rightarrow \mathbb{R}$ is said to *convex* if for any $t \in [0, 1]$ and any $x_1, x_2 \in (a, b)$, we have $f((1-t)x_1 + tx_2) \leq (1-t)f(x_1) + tf(x_2)$.
- (i) Show that a convex function on (a, b) is continuous. Give an example of a convex function that is not differentiable.
- (ii) Let $f: (a, b) \rightarrow \mathbb{R}$ be 2-times differentiable function. Show that f is convex on (a, b) if and only if $f''(x) \geq 0, \forall x \in (a, b)$.
- (iii) Using convexity of the exponential function, prove the generalized AM-GM inequality

$$x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n} \leq a_1 x_1 + a_2 x_2 + \cdots + a_n x_n,$$

where x_i and a_i ($1 \leq i \leq n$) are non-negative reals and $a_1 + a_2 + \cdots + a_n = 1$.

- Q6 (Optional) Consider the function $f(x) = \begin{cases} e^{-\frac{1}{x}} & \text{if } x > 0, \\ 0 & \text{if } x \leq 0. \end{cases}$

- (i) Show that f is continuous at $x = 0$. Hence f is continuous.
- (ii) Show that f is differentiable at 0 and $f'(0) = 0$. Hence, deduce that f is continuously differentiable and $f'(x) = \begin{cases} \frac{1}{x^2} e^{-\frac{1}{x}} & \text{if } x > 0, \\ 0 & \text{if } x \leq 0. \end{cases}$
- (iii) By induction, show that $f^{(m)}(x) = \begin{cases} P_m\left(\frac{1}{x}\right) e^{-\frac{1}{x}} & \text{if } x > 0, \\ 0 & \text{if } x \leq 0, \end{cases}$ where $P_m(t)$ is a polynomial in t of degree $2m$. Further, show that $f^{(m+1)}(0) = 0$. Deduce that f is smooth but not real analytic.